

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.2 CHEMICAL FAMILIES: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have spent eight lessons collecting evidence about chemical families. Now you will use it to answer the Driving Question: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm, and how do their family properties explain the materials that build, light up, and sustain our daily lives? Answer each section in the space below its prompt. Write in full sentences. Use the Claim-Evidence-Reasoning (CER) pattern where asked: make a claim, back it with evidence from our lessons, and explain your reasoning using electrons. Refer to the Thika Road street lamp, the Kamukunji welding shield and the helium balloon. Include balanced equations where the prompt asks for them.

Part 1: What is a chemical family?

Sodium, argon and helium are all elements, yet they behave very differently. (a) In your own words, what is a chemical family? (b) What single feature of an atom decides which family an element belongs to? (c) Name four families we studied in detail and give one element from each.

Write your answer here:

Part 2: Alkali metals and the Thika Road street lamp (CER)

Write a Claim-Evidence-Reasoning answer to this question: Why does sodium react violently with water, while lithium reacts less vigorously and potassium more vigorously? (a) Give a claim. (b) Give evidence from our water-reaction demonstration. (c) Explain your reasoning using valence electrons and shells. (d) Write the balanced equation for sodium reacting with water. (e) Sodium is also the element inside the Thika Road street lamp. What does this tell you about how sodium atoms can take in energy and give out light?

Write your answer here:

Part 3: Alkaline earth metals and Kenyan farms

Magnesium is in Group 2 and sodium is in Group 1. (a) Compare how magnesium and sodium reacted in our experiments. (b) Use the number of outer electrons and the energy needed to remove them to explain the difference. (c) Write the balanced equation for magnesium reacting with dilute hydrochloric acid. (d) Calcium is a Group 2 element, and CAN fertiliser used by Rift Valley maize farmers contains calcium. Explain why a Group 2 element can be useful and safe in a fertiliser bag stored in an agrovet shop, whereas sodium metal could not be.

Write your answer here:

Part 4: Noble gases and transition metals

(a) Explain why argon is used in the Kamukunji welding shield and helium in the party balloon, using the idea of outer shells. (b) Iron in a jembe and zinc on a mabati roof are transition metals. Describe how their reactivity compares with sodium and how this affects their use. (c) Write a balanced equation for one reaction of magnesium or sodium that we studied, and say which family the metal belongs to.

Write your answer here:

Part 5: Answering the Driving Question

Write a connected paragraph (6-8 sentences) answering the Driving Question: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm, and how do their family properties explain the materials that build, light up, and sustain our daily lives? Mention at least three families and one Kenyan example for each. Then state how your thinking has changed since Lesson 1.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of chemical family and valence electron ideas	Clearly and accurately links every family to its number of valence electrons and uses this to explain behaviour in every part.	Links most families to valence electrons correctly, with minor gaps or imprecise wording.	Names families or elements but does not connect behaviour to valence electrons.
Claim-Evidence-Reasoning structure	Gives a clear claim, specific evidence from the lessons (observations such as fizzing, pink phenolphthalein, squeaky pop) and reasoning that links the evidence to electron arrangement.	Gives a claim and some evidence, but the reasoning is partly missing or not clearly linked.	Gives statements without evidence or reasoning.
Balanced equations and chemical accuracy	All required equations are correct and balanced, and the products are correctly identified.	Equations are mostly correct with one small error in balancing or state.	Equations are missing or incorrect.

Connection to the anchoring phenomenon and Kenyan contexts	Explains the Thika Road lamp, Kamukunji welding and helium balloon, plus farm and hardware examples, using family properties.	Explains most of the phenomenon examples with some Kenyan contexts.	Mentions few examples or does not link them to chemical families.
Clarity and growth in understanding	Writes a well-organised, connected paragraph and clearly shows how their ideas changed since Lesson 1.	Writes a mostly clear explanation with a brief note on how their ideas changed.	Writing is disconnected and does not show how their ideas changed.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
